Linear Algebra for Quantum Computing

Linear Maps, Eigenvalues, Tensor Products

Quantum Computing

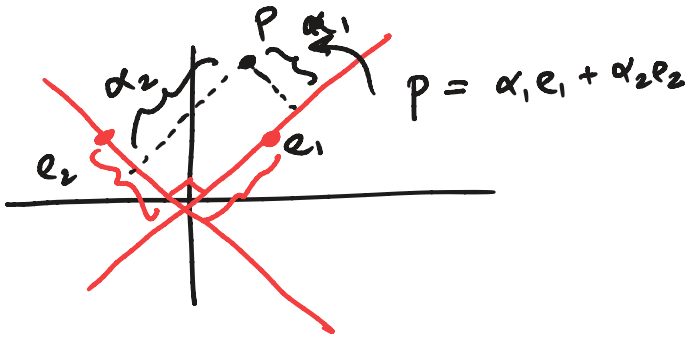
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Recap: What We Already Have

$$\begin{bmatrix} \bar{u}_1 & \dots & \bar{u}_n \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$



- Vectors $\mathbf{v} \in \mathbb{C}^n$ (bold); scalars in $\mathbb{C}$
- $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\dagger \mathbf{v} = \sum_j \overline{u_j} v_j$
- ONB $\{\mathbf{e}_i\}$: $\mathbf{v} = \sum_i \langle \mathbf{e}_i, \mathbf{v} \rangle \, \mathbf{e}_i$

Bases of $\mathbb{C}^2$

$\mathbf{e}_1, \mathbf{e}_2$ (standard)

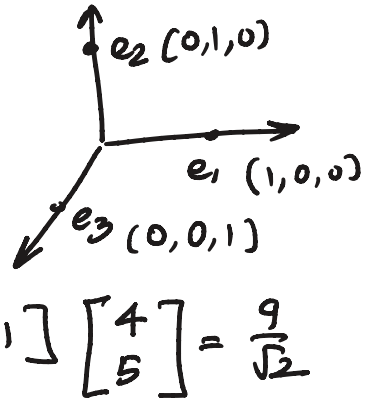
$\mathbf{h}_\pm = \frac{1}{\sqrt{2}}(1, \pm 1)^T$ (Hadamard)

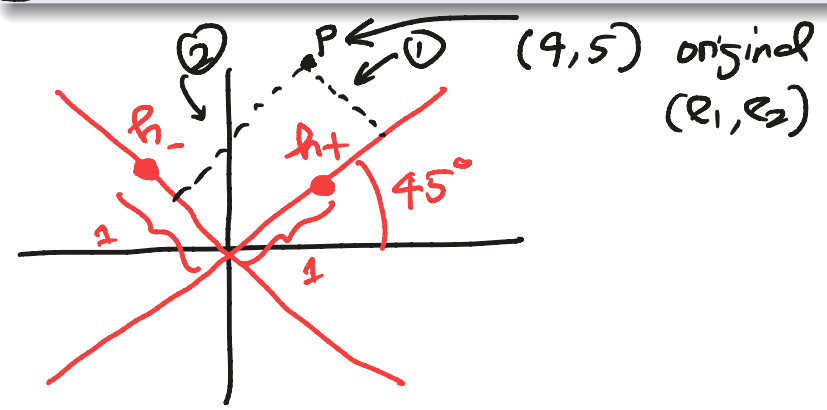
$$\bar{u}_1 \cdot v_1 + \dots + \bar{u}_n \cdot v_n$$

$p = 4e_1 + 5e_2$

$p = \alpha_1 h_+ + \alpha_2 h_-$

$\alpha_1 = \langle h_+, p \rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \frac{9}{\sqrt{2}}$





A Single Qubit Is a Vector in $\mathbb{C}^2$

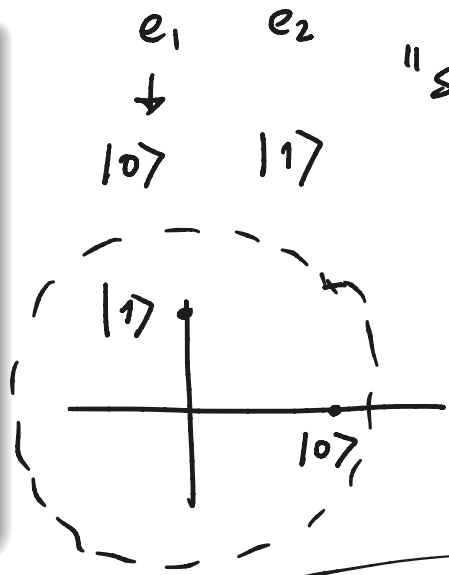
Classical bit vs qubit

Bit: exactly one of 0 or 1.

Qubit: a **unit vector**

$$\psi = \alpha\,\mathbf{e}_1 + \beta\,\mathbf{e}_2 = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$$

with $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$.



"superposition"

$|\psi\rangle$ $\quad |0\rangle \to \begin{bmatrix}1\\0\end{bmatrix}$ $\quad \alpha|0\rangle + \beta|1\rangle$

$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

$|0\rangle = 1\cdot|0\rangle + 0\cdot|1\rangle$

$|0\rangle + |1\rangle$? No

A Single Qubit Is a Vector in $\mathbb{C}^2$

Classical bit vs qubit

Bit: exactly one of 0 or 1.

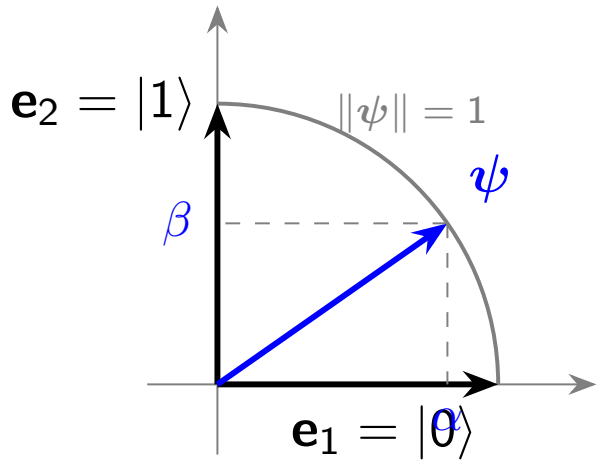
Qubit: a **unit vector**

$$\psi = \alpha\,\mathbf{e}_1 + \beta\,\mathbf{e}_2 = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$$

with $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$.

- The basis vectors play the role of the bit values: $\mathbf{e}_1 = |0\rangle$, $\mathbf{e}_2 = |1\rangle$.
- α, β are **amplitudes**; measuring gives 0 w.p. $|\alpha|^2$, 1 w.p. $|\beta|^2$.

"Unit vector" = "the probabilities add up to 1".



real cartoon: in general α, β are complex

Today's Plan

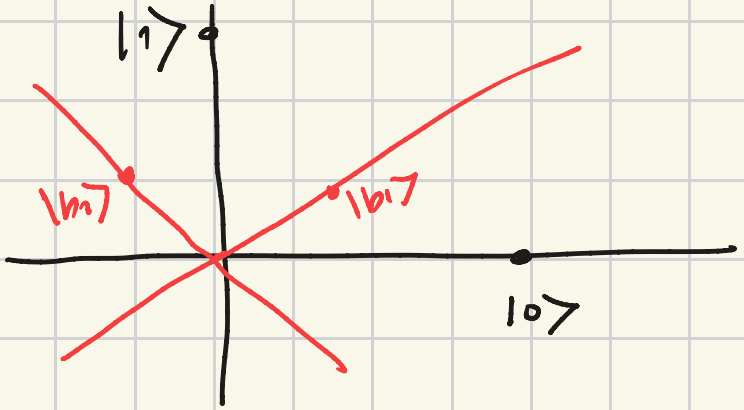
Matrices
Linear Maps

Hermitian, unitary, projectors

Tensor Product

Eigenvalues
Eigenvectors

- Same notation as last time: bold vectors, $\langle \mathbf{u}, \mathbf{v} \rangle$, $A^\dagger$.



$|0\rangle \quad |1\rangle \qquad\qquad |\psi\rangle$

$|b_1\rangle, |b_2\rangle$

$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

outcome?

$|0\rangle$ $\frac{1}{2}$ prob

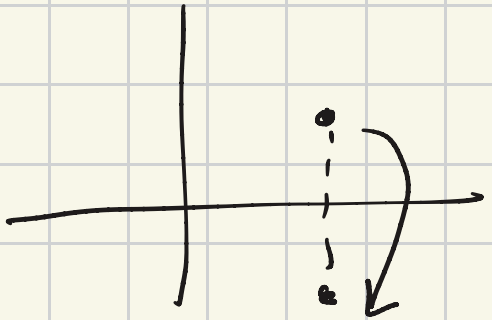
$|1\rangle$ $\frac{1}{2}$ prob.

"Measurement" ? $|\psi\rangle$ Qubit in this basis.

$|\psi\rangle = \alpha_1|b_1\rangle + \alpha_2|b_2\rangle$ $\qquad |b_1\rangle$ OR $|b_2\rangle$

$|\alpha_1|^2 \qquad |\alpha_2|^2$

↑ prob. $|b_1\rangle$



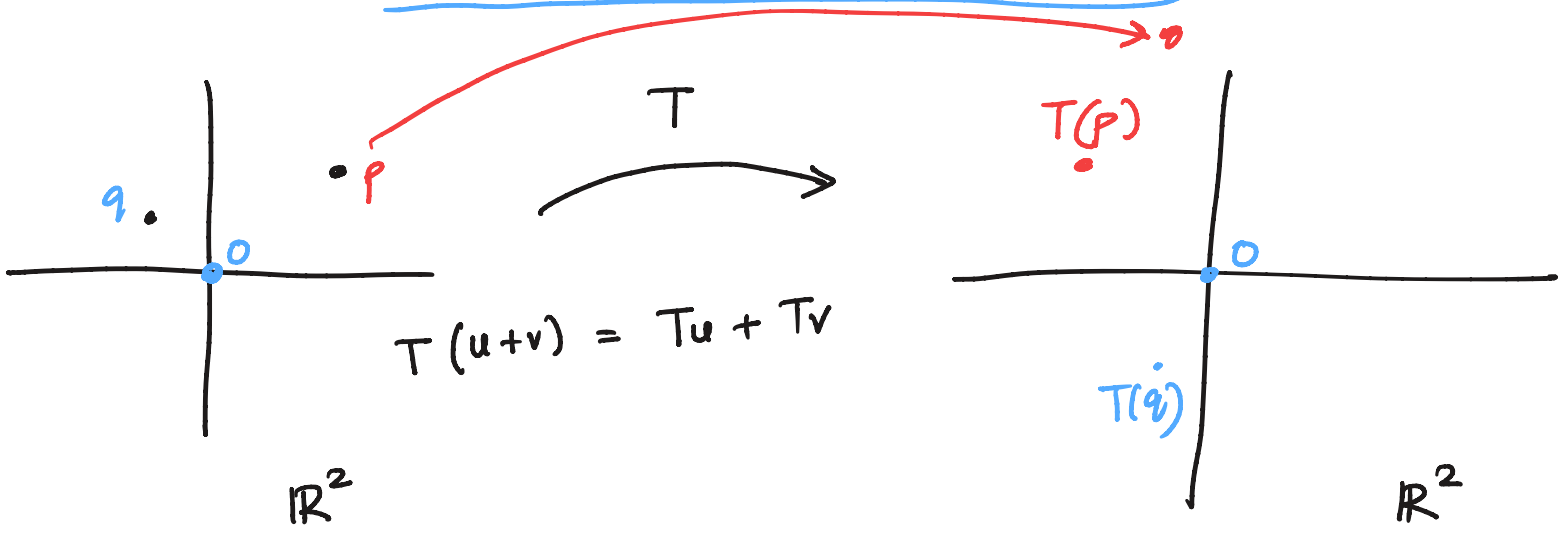
Linear Maps

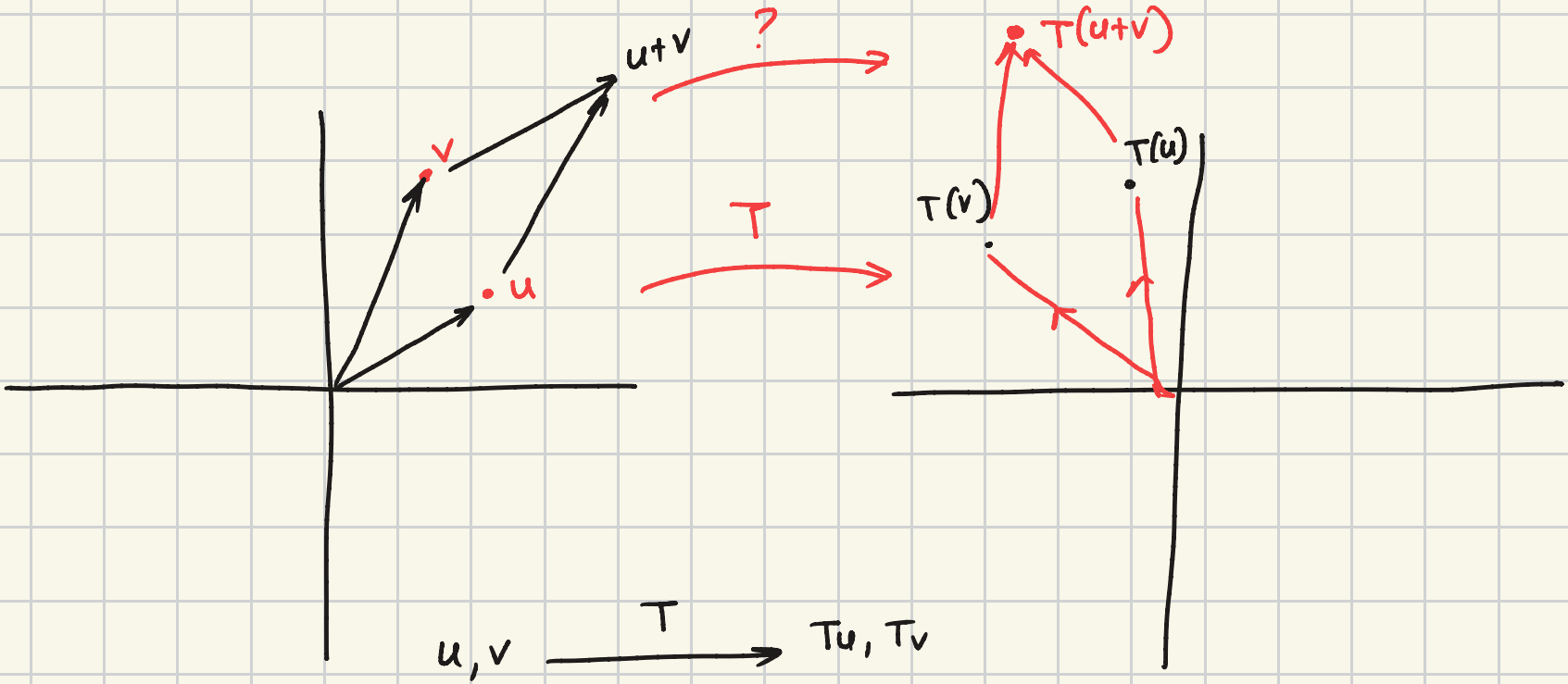
Definition

$T : V \to W$ is **linear** if for all $\mathbf{u}, \mathbf{v}$ and $\alpha, \beta \in \mathbb{C}$

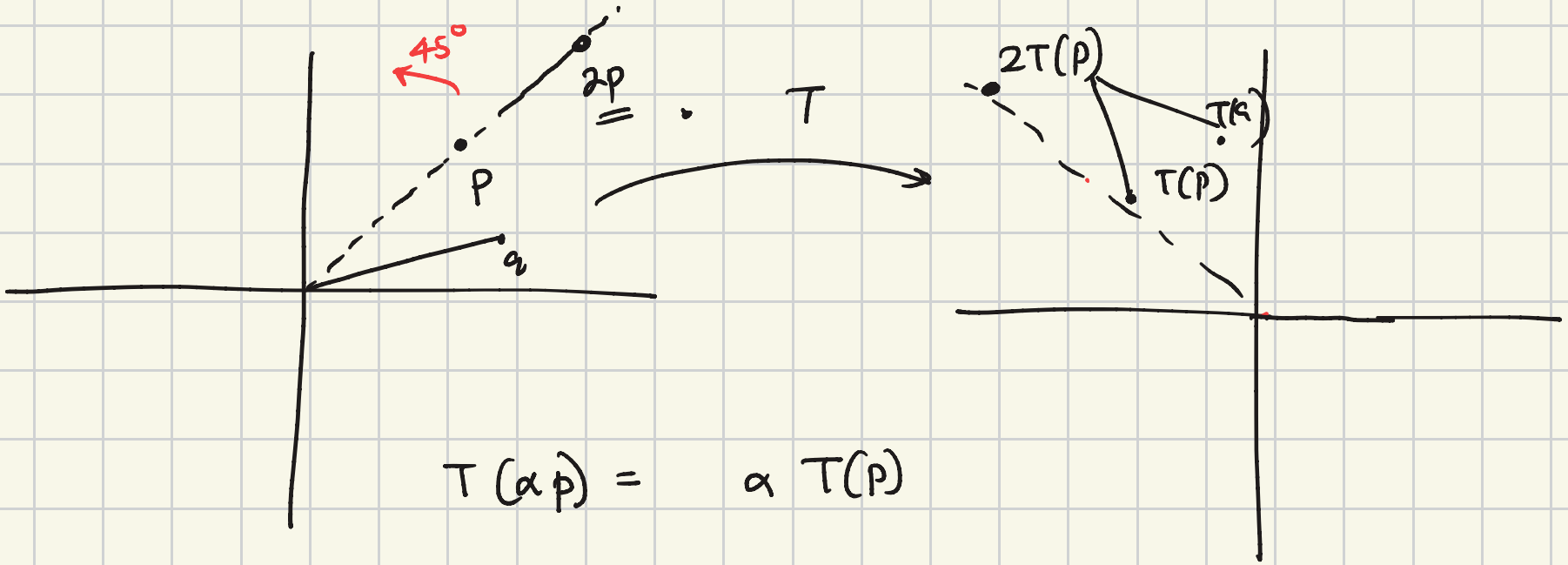
$\mathbb{C}^2 \quad \mathbb{C}^2$

$$T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha\, T(\mathbf{u}) + \beta\, T(\mathbf{v}).$$

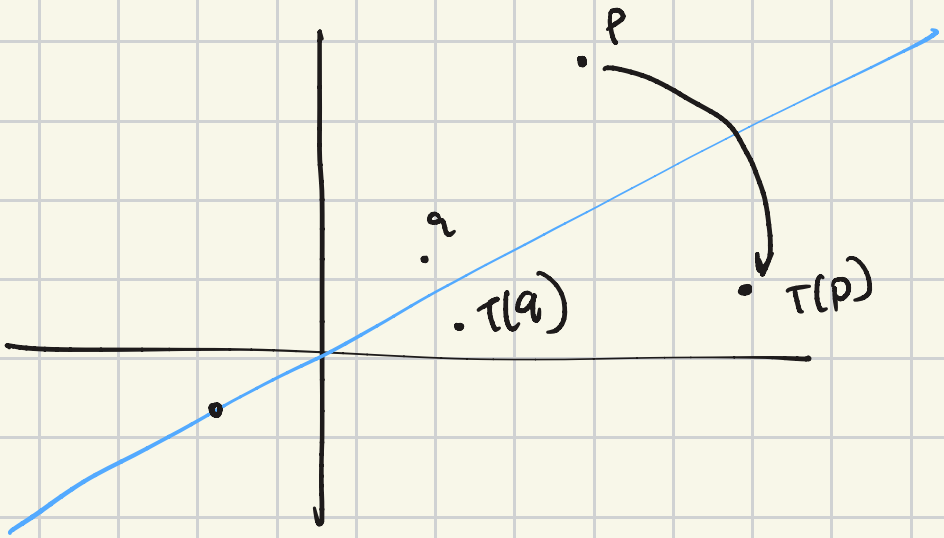




$$
\begin{array}{ccc}
u, v & \xrightarrow{T} & Tu, Tv \\
\downarrow + & & \downarrow + \\
u+v & \xrightarrow{T} & Tu + Tv = T(u+v)
\end{array}
$$



$$T(\alpha p) = \alpha\, T(p)$$


p
q
T(q)
T(p)

Linear Maps

Definition

$T : V \to W$ is **linear** if for all $\mathbf{u}, \mathbf{v}$ and $\alpha, \beta \in \mathbb{C}$

$$T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha\, T(\mathbf{u}) + \beta\, T(\mathbf{v}).$$

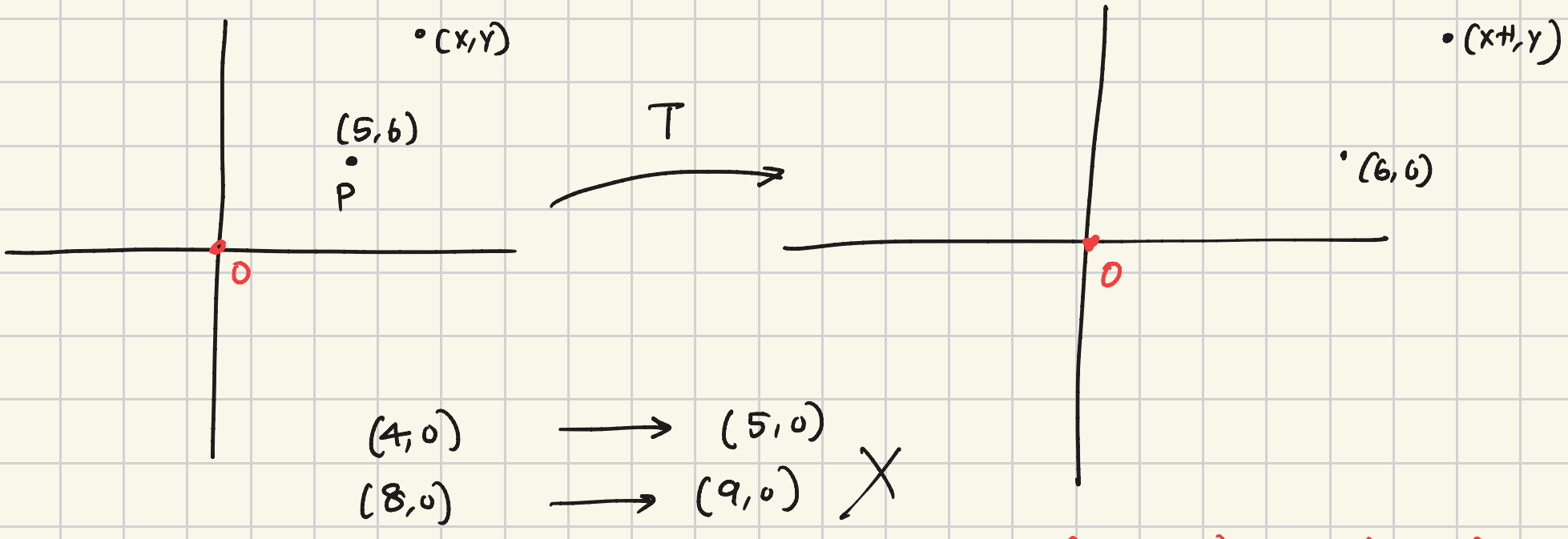
Linear ✓

- rotations, reflections, scalings
- $\mathbf{v} \mapsto A\mathbf{v}$ for a matrix A
- every quantum gate

Not linear ✗

- $\mathbf{v} \mapsto \mathbf{v} + \mathbf{b}$ $(\mathbf{b} \neq \mathbf{0})$: moves $\mathbf{0}$
- $\mathbf{v} \mapsto \|\mathbf{v}\|$
- $\mathbf{v} \mapsto \overline{\mathbf{v}}$: $T(i\mathbf{v}) = -i\,\overline{\mathbf{v}} \neq i\, T(\mathbf{v})$

Linearity is why a gate applied to a superposition acts on each term separately.



$(4,0) \longrightarrow (5,0)$

$(8,0) \longrightarrow (9,0)$ ✗

$T(0) = 0$

$T(\alpha 0) = \alpha T(0)$ ← linearity

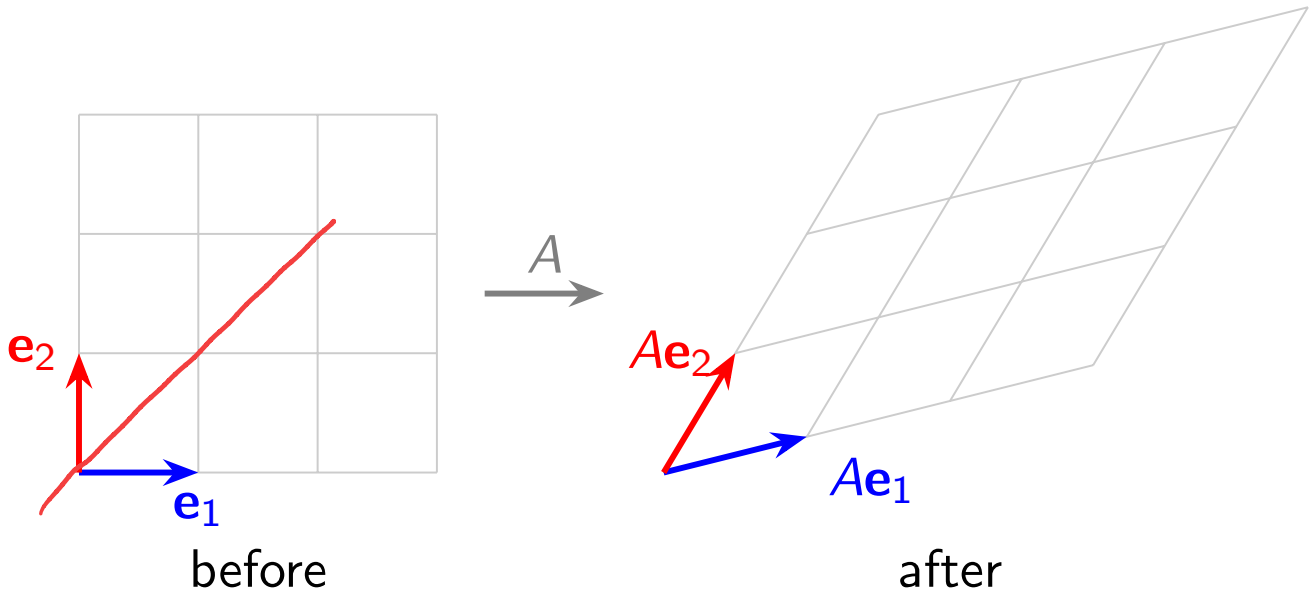
$T(\underbrace{2(0,\ldots,0)}) = 2\,T(0,\ldots,0)$

$= T(0,\ldots,0)$

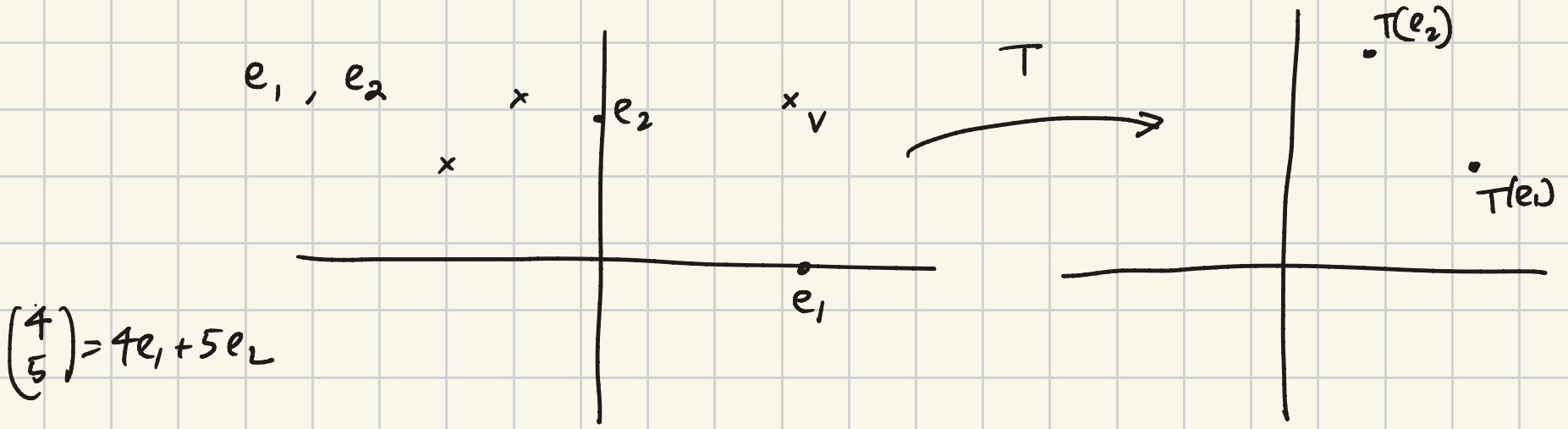
What a Linear Map Does to the Plane

- $\mathbf{0}$ stays at $\mathbf{0}$.
- Lines stay lines; grid stays evenly spaced.
- Everything is decided by where $\mathbf{e}_1, \mathbf{e}_2$ go.

Here $A = \begin{pmatrix} 1.2 & 0.6 \\ 0.3 & 1 \end{pmatrix}$.



How do we construct linear maps in general?



$\begin{pmatrix} 4 \\ 5 \end{pmatrix} = 4e_1 + 5e_2$

If we know $T(e_1)$ and $T(e_2)$, then we know $T(v)$ for any vector v

$$T\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \qquad T = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$T\begin{bmatrix} 4 \\ 5 \end{bmatrix} = T\left[\underbrace{4\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{u} + \underbrace{5\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{v} \right] = T\left[4\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right] + T\left[5\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]$$

$$T\begin{bmatrix}4\\5\end{bmatrix} = 4\,T\begin{bmatrix}1\\0\end{bmatrix} + 5\,T\begin{bmatrix}0\\1\end{bmatrix}$$

$$= 4\begin{bmatrix}4\\5\end{bmatrix} + 5\begin{bmatrix}1\\-3\end{bmatrix}$$

$$T\begin{bmatrix}x\\y\end{bmatrix} = x\begin{bmatrix}4\\5\end{bmatrix} + y\begin{bmatrix}1\\-3\end{bmatrix}$$

$$= \begin{bmatrix}4 & 1\\5 & -3\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}$$

(M)

T

$|0\rangle \longrightarrow \dfrac{|0\rangle+|1\rangle}{\sqrt{2}}$

$|1\rangle \longrightarrow \dfrac{|0\rangle-|1\rangle}{\sqrt{2}}$

$\alpha|0\rangle+\beta|1\rangle \xrightarrow{T} \alpha T|0\rangle + \beta T|1\rangle.$

1st 2nd

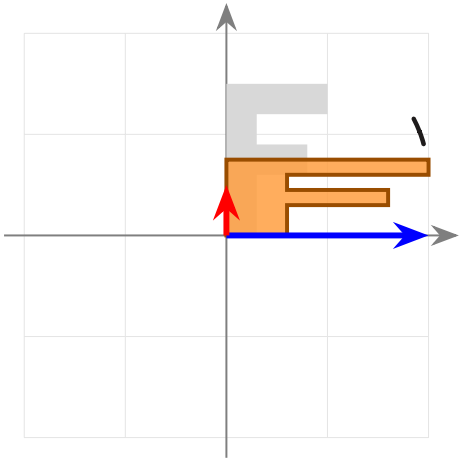
$$M = \begin{bmatrix}3 & 1\\1 & i\end{bmatrix}\begin{pmatrix}x\\y\end{pmatrix}$$

$$\begin{pmatrix}1\\0\end{pmatrix} \longrightarrow \begin{pmatrix}3\\1\end{pmatrix}$$

$$\begin{pmatrix}0\\1\end{pmatrix} \longrightarrow \begin{pmatrix}1\\i\end{pmatrix}$$

A Gallery of Linear Maps (1)

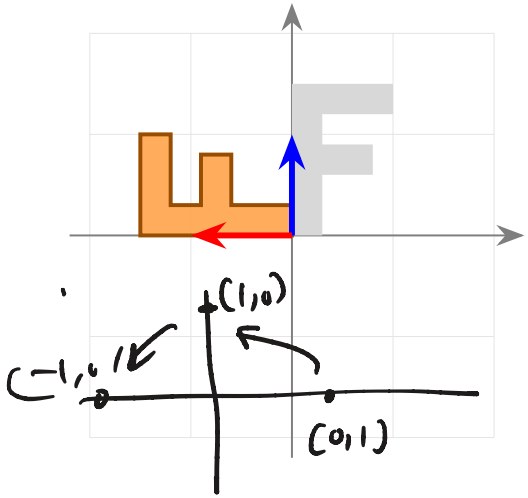
Grey: original shape. Orange: its image. Blue/red arrows: images of $\mathbf{e}_1$, $\mathbf{e}_2$ = the columns of the matrix.



Scaling

$$\begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

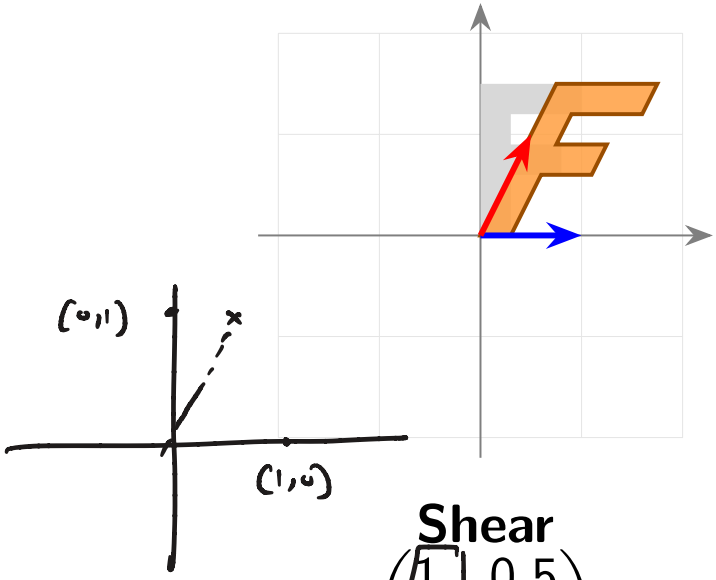
stretch x, squash y



Rotation by $90°$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

counter-clockwise

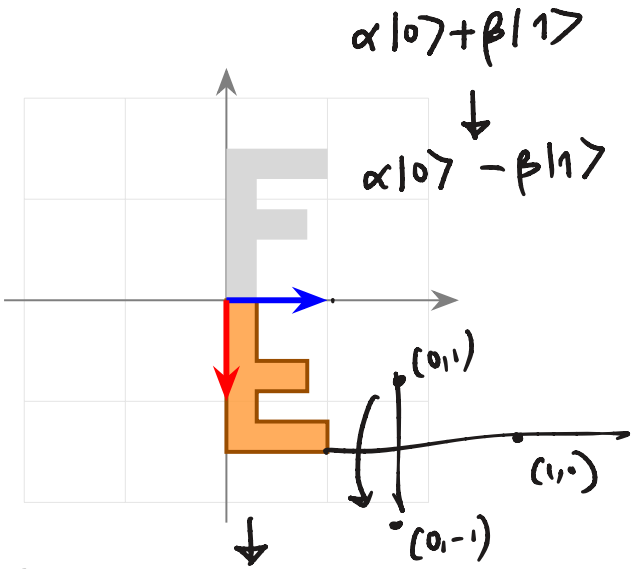


Shear

$$\begin{pmatrix} 1 & 0.5 \\ 0 & 1 \end{pmatrix}$$

slide rows sideways

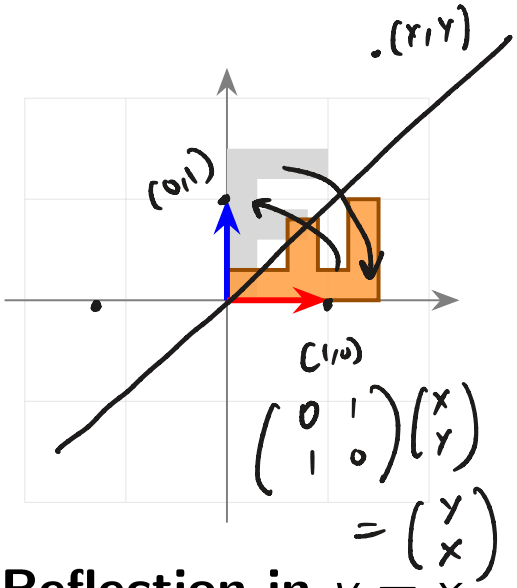
A Gallery of Linear Maps (2)



Reflection in the x-axis

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

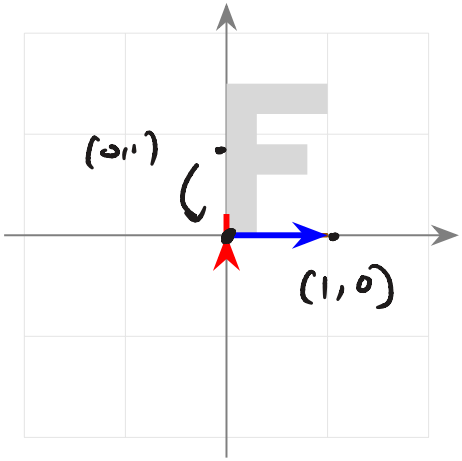
the F is now mirror-imaged



Reflection in $y = x$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

swaps the two axes



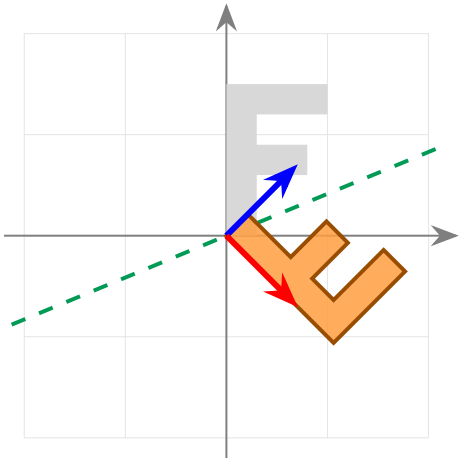
Projection onto x-axis

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

flattened: not invertible

Reflections and rotations keep lengths and angles (unitary); scaling, shear and projection do not.

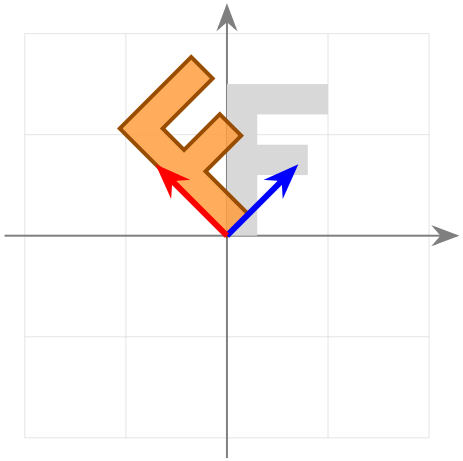
A Gallery of Linear Maps (3)



Hadamard

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

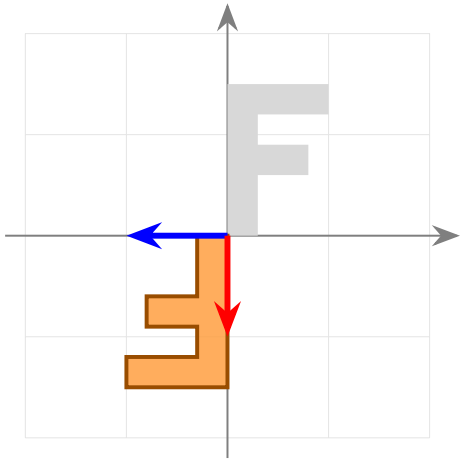
reflection in the line at 22.5° (dashed)



Rotation by 45°

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

general: R_θ, columns $(\cos\theta, \sin\theta)$, $(-\sin\theta, \cos\theta)$



Minus the identity

$$-I = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

rotation by 180°, *not* a reflection

From Linear Map to Matrix

Write $\mathbf{v} = \sum_j v_j \mathbf{e}_j$. By linearity

$$T(\mathbf{v}) = \sum_j v_j \, T(\mathbf{e}_j).$$

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

$$v_1 \, T(e_1) + v_2 T(e_2) + \cdots$$

From Linear Map to Matrix

Write $\mathbf{v} = \sum_j v_j \mathbf{e}_j$. By linearity

$$T(\mathbf{v}) = \sum_j v_j \, T(\mathbf{e}_j).$$

So T is fixed by the n vectors $T(\mathbf{e}_1), \ldots, T(\mathbf{e}_n)$. Stack them as columns:

$$[T] = \Big(\; T(\mathbf{e}_1) \;\Big|\; T(\mathbf{e}_2) \;\Big|\; \cdots \;\Big|\; T(\mathbf{e}_n) \; \Big), \qquad T(\mathbf{v}) = [T]\,\mathbf{v}.$$

[T]

From Linear Map to Matrix

Write $\mathbf{v} = \sum_j v_j \mathbf{e}_j$. By linearity

$$T(\mathbf{v}) = \sum_j v_j \, T(\mathbf{e}_j).$$

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$$[T] = \Big(\; T(\mathbf{e}_1) \;\Big|\; T(\mathbf{e}_2) \;\Big|\; \cdots \;\Big|\; T(\mathbf{e}_n) \; \Big), \qquad T(\mathbf{v}) = [T]\,\mathbf{v}.$$

Rule

Column j of the matrix = image of the j-th basis vector.

This is the single most useful recipe for writing down gate matrices.

Important Examples

Bit flip X "operator"

$X\mathbf{e}_1 = \mathbf{e}_2, \quad X\mathbf{e}_2 = \mathbf{e}_1$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \to \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \to \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$\alpha|0\rangle + \beta|1\rangle \to \alpha|1\rangle + \beta|0\rangle$

Phase flip Z

$Z\mathbf{e}_1 = \mathbf{e}_1, \quad Z\mathbf{e}_2 = -\mathbf{e}_2$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$$

$\alpha|0\rangle + \beta|1\rangle$

$\alpha|0\rangle - \beta|1\rangle.$

$$H \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} x+y \\ x-y \end{pmatrix}$$

Hadamard H

$H\mathbf{e}_1 = \mathbf{h}_+, \quad H\mathbf{e}_2 = \mathbf{h}_-$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \to \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad x$$

h_+

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \to \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad y$$

h_-

Important Examples

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$X\mathbf{e}_1 = \mathbf{e}_2, \;\; X\mathbf{e}_2 = \mathbf{e}_1$

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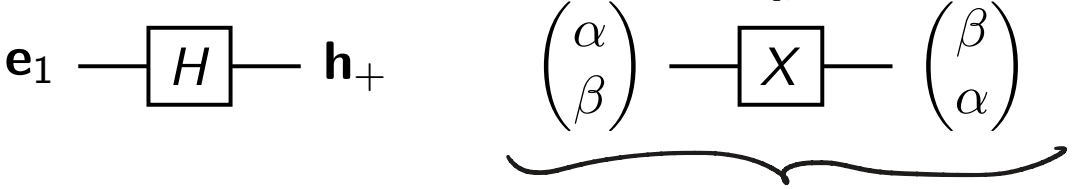
Hadamard H

$H\mathbf{e}_1 = \mathbf{h}_+, \;\; H\mathbf{e}_2 = \mathbf{h}_-$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

On a general state, by linearity:

$$X\left(\alpha\,\mathbf{e}_1 + \beta\,\mathbf{e}_2\right) = \alpha\,\mathbf{e}_2 + \beta\,\mathbf{e}_1 = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}, \qquad Z\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ -\beta \end{pmatrix}.$$



A wire in a circuit carries a vector; a box applies a matrix to it.

Matrix Entries as Inner Products

With an ONB $\{\mathbf{e}_i\}$:

Entries

$$A_{ij} = \langle \mathbf{e}_i,\, A\mathbf{e}_j \rangle = \mathbf{e}_i^\dagger A\, \mathbf{e}_j \qquad (\text{row } i,\ \text{column } j)$$

Matrix Entries as Inner Products

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Reconstruction from outer products

$$A = IAI = \sum_{i,j} \mathbf{e}_i \left(\mathbf{e}_i^\dagger A\, \mathbf{e}_j\right) \mathbf{e}_j^\dagger = \sum_{i,j} A_{ij}\, \mathbf{e}_i \mathbf{e}_j^\dagger$$

Matrix Entries as Inner Products

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Examples

$$X = \mathbf{e}_1\mathbf{e}_2^\dagger + \mathbf{e}_2\mathbf{e}_1^\dagger, \qquad Z = \mathbf{e}_1\mathbf{e}_1^\dagger - \mathbf{e}_2\mathbf{e}_2^\dagger, \qquad I = \mathbf{e}_1\mathbf{e}_1^\dagger + \mathbf{e}_2\mathbf{e}_2^\dagger.$$